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From Constituents to Index: Interpretable Price Movement Prediction via Cross-Asset Order Flow

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Abstract

This study proposes a novel deep learning architecture for intraday index price movement prediction, designed to capture the spatio-temporal dependencies between a market index and its constituents. Our interpretable, multi-stage framework addresses the limitations of single-asset black box models by: (i) dynamically selecting the 30 most influential constituents, (ii) forecasting each selected asset's future order flow imbalance(OFI) sequence, and (iii) fusing the predicted OFI tensor through a hierarchical Stock-Time Attention block to classify the short-term index direction. Extensive experiments on KOSPI 200 data demonstrate that our model significantly outperforms existing baselines. The results confirm that incorporating cross-asset information, particularly from constituent futures, is critical for enhancing forecast accuracy. We further validate our framework's interpretability by showing that the learned attention weights effectively identify primary market drivers. Assets with high attention scores exhibit price patterns that are quantitatively more similar to the aggregate index.

CCS Concepts

• **Applied computing** → *Forecasting*; • **Computing methodologies** → *Machine learning*.

Keywords

Limit Order Book, Attention Mechanism, Market Microstructure, Order Flow Imbalance, Financial Market Prediction

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1 Introduction

Predicting future movements of financial markets has been a core challenge in both academic research and industry practice. Early studies have relied on econometric approaches, such as ARIMA and Markov models, to capture market dynamics [1, 7, 9, 19]. While these approaches yield valuable insights, their effectiveness is limited by strict statistical assumptions and may struggle to capture the complex, non-linear dynamics portrayed in modern financial markets.

The advent of advanced computing technologies and the availability of high-frequency data have shifted the research focus toward short-term price prediction. In particular, the Limit Order Book (LOB) has become a focal point of research. By providing a granular, real-time representation of market depth, and supply-demand dynamics, it serves as a crucial data source for short-term price prediction. Consequently, deep learning models have emerged as a powerful tool for learning the complex patterns within such high-frequency data [3, 13–16, 20]. Nonetheless, applying these models directly to raw LOB data presents substantial challenges due to its high dimensionality, inherent noise, and complex temporal dependencies. To address these specific issues, we engineer a more condensed and potent feature, Order Flow Imbalance (OFI) [5, 6], to represent net market pressure instead of using the raw data. This approach effectively mitigates the aforementioned difficulties, as detailed in Section 3.1.

Beyond the issue of data representation, the prevailing deep learning approaches themselves carry two critical limitations. First, most studies concentrate on predicting the price movements of a single-asset independently, overlooking the complex interdependencies among multiple assets. However, financial markets are interconnected systems where assets influence each other significantly. Ignoring these relationships reduces predictive accuracy and limits our understanding of market dynamics. Second, deep learning models typically operate as black boxes, making their decision-making processes difficult to interpret. This lack of transparency is problematic for practitioners and regulators who need to understand which assets drive market changes and how assets influence each other over time.

To address these limitations, we propose an interpretable, multi-stage predictive framework designed explicitly for modeling financial markets as interconnected systems.

Directly incorporating data from numerous constituent stocks is computationally expensive and noisy. Therefore, we use Variable Selection Network (VSN) [11] to dynamically rank and select a subset of the most influential stocks. The selected asset data are then processed by our main predictive pipeline. In the first stage, we employ the Informer model [21] to generate forward-looking predictive features for each selected asset. This creates a rich tensor representing anticipated future movements, which is then fed into our Spatio-Temporal Classification Network. This network pursues two primary objectives. First, it captures the collective impact of constituent stocks using a hierarchical Stock-Time Attention mechanism that evaluates cross-asset relationships and their temporal evolution. Second, it resolves the black box problem; analyzing the model's attention scores reveals key driving assets and their lead-lag relationships, providing a transparent analytical tool.

This integrated approach provides significant predictive advantages and clear interpretability of market dynamics. The main contributions of this study are:

(1) We propose a novel model architecture explicitly designed to capture the spatio-temporal dependencies between a market index and its constituents, treating the market as an interconnected system.
(2) Through extensive experiments, we demonstrate that our proposed model outperforms state-of-the-art single-asset predictive models, highlighting the benefits of incorporating cross-asset information.
(3) Our framework provides robust interpretability. By analyzing the model's internal mechanisms, we can identify which constituent assets are the primary drivers of index movements and infer the lead-lag relationships between the index and its components, offering valuable insights into market microstructure.

2 Related Works

OFI is recognized as a crucial feature for explaining short-term price fluctuations, with its effectiveness validated across numerous studies. A pioneering study first established a strong linear relationship between OFI at the best bid and ask prices and subsequent price movements, identifying OFI as a fundamental driver in price formation [6]. This relationship was subsequently reaffirmed in the German stock market, where it was also shown to be related to market microstructure characteristics such as company size and liquidity [8]. The concept was further extended to multi-asset environments, revealing that the past OFI of one asset provides significant predictive power for the future returns of other assets [5]. Recent deep learning research has also highlighted the value of OFI; extensive experiments have shown that models utilizing OFI as an input feature significantly outperform those based solely on raw LOB data [10]. This underscores OFI's capability to effectively summarize microstructural market trends.

While OFI provides a powerful summary, much research has focused on applying deep learning directly to raw LOB data. An early seminal study demonstrated the superiority of Convolutional Neural Networks (CNNs) over other methodologies such as Support Vector Machines (SVMs) and Multi-Layer Perceptrons (MLPs) [16]. Subsequently, the DeepLOB model was introduced, a hybrid model combining CNNs to capture the spatial structure of the LOB and Long Short-Term Memory (LSTM) networks to model temporal dependencies [20]. Following this, efforts to improve efficiency and interpretability led to the development of models like the Temporal Attention-Augmented Bilinear network (TABL), which showed that a shallow structure with a temporal attention mechanism could outperform deeper models [14]. This was further enhanced with the Bilinear Normalization (BiN) method to address data non-stationarity [15]. Despite these advances, an extensive benchmarking study highlighted that many state-of-the-art models exhibit significant overfitting, resulting in poor generalization to new market data and underscoring a key challenge in the field [13]. To address these challenges of generalization and long-range dependency modeling, recent research has increasingly turned to the Transformer architecture. A notable example in the LOB domain is the TLOB model, which incorporates a dual attention mechanism to simultaneously capture temporal and spatial (feature-level) dependencies, leading to improved performance in long-term prediction tasks [3]. The success of such models stems from the foundational principles of the Transformer, first introduced to capture long-range dependencies in sequential data exclusively through self-attention mechanisms [17]. However, applying the original Transformer to long time series presents challenges, leading to several key innovations. To tackle the $O(L^2)$ computational complexity, the Informer model was proposed, which reduces complexity to $O(L \log L)$ with ProbSparse attention [21]. Furthermore, to enhance interpretability and integrate diverse variables, the Temporal Fusion Transformer (TFT) was developed, a hybrid architecture that effectively combines complex inputs like static variables and known future inputs, making it a powerful and interpretable tool for time series forecasting [11].

3 Model

In this section, we describe the proposed model. The core architecture, presented in Figure 1, is composed of two main stages: Stage 1, an Informer-based module that predicts OFI, and Stage 2, an attention network that processes the predicted OFI using Stock-wise and Time-wise Attention.

3.1 Input Feature Engineering: Order Flow Imbalance

The foundational data for our analysis is the raw LOB. To capture the market's temporal dynamics, the initial input to our model is a matrix X_t containing the T most recent LOB states. This input is defined as $X_t = [x_{t-T+1}, \ldots, x_t]^T \in \mathbb{R}^{T \times 20}$, where each row x_t represents the LOB state at time t. The 20-dimensional feature vector x_t is constructed from the top five price levels as follows:

$$x_t = [p_{\text{ask}}^{(i)}(t), v_{\text{ask}}^{(i)}(t), p_{\text{bid}}^{(i)}(t), v_{\text{bid}}^{(i)}(t)]_{i=1}^{5}.$$

However, predicting short-term price movements from this raw LOB data is a non-trivial task. While the LOB provides a complete, tick by tick record of market activity, this granularity introduces significant informational overhead. For the task of mid-price prediction, much of this data can act as noise, increasing the risk of the model overfitting to spurious, transient patterns rather than learning the underlying directional signals.

To address this, we shift from using raw data to a more abstract and potent feature: OFI. Instead of processing every individual order event, OFI aggregates order flows over a time interval. This process effectively distills the high-frequency chaos of the LOB into a condensed representation of the market's net buying or selling pressure. This condensed feature has been proven highly effective for short-term price prediction in numerous studies [5, 6, 8, 10, 18].

For our framework, we compute OFI features for our entire universe of assets. The initial input to our model is therefore a comprehensive feature set comprising the OFIs of two main sources: (1) the market index futures and (2) all constituent stocks under consideration. This creates the high-dimensional, multi-asset input that our dynamic selection stage is designed to process.

The OFI at the i^{th} price level over a time interval h is defined as follows. At time t, it is the sum of net order flow over all individual order book update events n within the preceding interval $(t-h, t]$:

$$\mathrm{OFI}_{i,t}^{m,h} = \sum_{n=N(t-h)+1}^{N(t)} \left(\mathrm{OF}_{i,n}^{m,b} - \mathrm{OF}_{i,n}^{m,a}\right) \tag{1}$$

Here, the terms $\mathrm{OF}_{i,n}^{m,b}$ (bid) and $\mathrm{OF}_{i,n}^{m,a}$ (ask) represent the order flow at each event n. They are calculated by tracking changes in quoted volume q at a given price level p, capturing new orders, cancellations, and modifications:

$$\mathrm{OF}_{i,n}^{m,b} = \begin{cases} q_{i,n}^{m,b} & \text{if } p_{i,n}^{m,b} > p_{i,n-1}^{m,b} \\ q_{i,n}^{m,b} - q_{i,n-1}^{m,b} & \text{if } p_{i,n}^{m,b} = p_{i,n-1}^{m,b} \\ 0 & \text{if } p_{i,n}^{m,b} < p_{i,n-1}^{m,b} \end{cases} \tag{2}$$

$$\mathrm{OF}_{i,n}^{m,a} = \begin{cases} q_{i,n}^{m,a} & \text{if } p_{i,n}^{m,a} < p_{i,n-1}^{m,a} \\ q_{i,n}^{m,a} - q_{i,n-1}^{m,a} & \text{if } p_{i,n}^{m,a} = p_{i,n-1}^{m,a} \\ 0 & \text{if } p_{i,n}^{m,a} > p_{i,n-1}^{m,a} \end{cases} \tag{3}$$

In this study, we compute OFIs for the first five levels of the order book ($m = 1, \ldots, 5$) for every asset in our initial dataset. This process yields a rich, multi-dimensional feature set that captures market depth. For the initial selection stage, however, we exclusively use the Level 1 OFI. Its lower dimensionality ensures faster computation times; a critical factor when processing a large pool of assets, while still providing an efficient and powerful signal for identifying key assets. Once the key tickers are identified, we feed the full five-level OFI data of those assets into the main forecasting model.

3.2 Stage 0: Dynamic Ticker Selection for Key Feature Extraction

Accurately predicting the movements of a financial index requires comprehensive consideration of numerous constituent assets. However, incorporating the time series data of hundreds of constituent assets directly as model inputs can introduce practical limitations such as overfitting, noise amplification, and high computational cost. Thus, it is crucial to effectively identify and select key driving tickers that influence the index movements.

We employ a Variable Selection Network (VSN) [11] to dynamically identify the 30 most influential constituent assets at each time step. This data-driven approach avoids the limitations of static

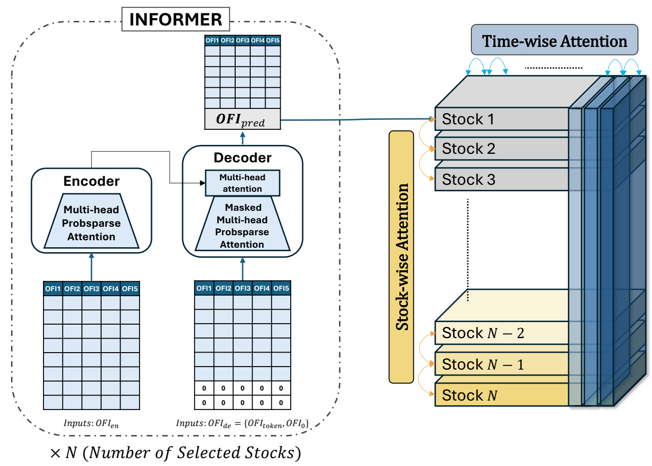


Figure 1: The core architecture of the proposed model. (Left) An Informer-based prediction module that generates future OFI (OFI_{pred}) sequences for N individual stocks. (Right) A hierarchical attention network that receives the predicted OFI tensor from the Informer and extracts spatio-temporal features by sequentially applying Stock-wise and Time-wise Attention.

methods and allows the model to focus only on key market drivers, reducing noise and computational cost.

The input to this network at each time step t is a vector Ξ_t, which consists of the Level 1 OFI values from all constituent assets. We use only the Level 1 OFI for this stage, as it represents the most immediate market pressure and has been shown to contain the strongest predictive signal for short-term price movements [6]. Specifically, VSN receives an input vector Ξ_t and computes an importance weight $v_{\chi_t}^{(j)}$ for each asset j via a Gated Residual Network (GRN) and a Softmax function: $v_{\chi_t} = \mathrm{Softmax}(\mathrm{GRN}_{v_\chi}(\Xi_t, c_s))$. These weights adapt based on each data instance, allowing the model to autonomously identify the variables most relevant to particular market conditions. Based on these learned weights, we rank all constituent assets by their importance and select the top 30 tickers. The time series data from only these selected key assets are then used as the final input for the main prediction model in the subsequent stages.

The GRN enables this dynamic selection through its unique structure. It employs a gating mechanism, the GLU, along with a skip connection. This allows the model to learn complex non-linear relationships while adaptively skipping transformations deemed irrelevant for a given input. This capability to filter out unnecessary complexity is crucial for enhancing robustness when dealing with highly volatile and noisy financial time series data.

In summary, this preliminary stage leverages VSN and GRN to intelligently quantify the predictive relevance of each asset in a large pool of constituent stocks. By selecting the top 30 assets with the highest importance weights, we construct a refined and focused input set. This process not only minimizes noise and computational burden but also maximizes the overall predictive performance of the subsequent time series forecasting model.

3.3 Stage 1: Informer-based Predictive Feature Generation

The first stage of our main predictive pipeline is to generate forward looking features for each of the selected assets. Instead of relying on retrospective statistical summaries, we employ a forecasting model to explicitly predict future market dynamics. Specifically, the model's objective is to predict a sequence of future OFI values for each individual asset over a given prediction horizon. In essence, this stage involves independently forecasting the future order flow of each key stock. For instance, if the prediction horizon is 5 seconds, the model will forecast the five subsequent OFI values for each selected stock.

These predicted OFI sequences for all selected assets are then collected and aligned into a unified multi-dimensional tensor. This tensor, which encapsulates the anticipated future supply and demand pressures of each individual asset, serves as the foundational input for the subsequent stage. It is in this next stage that the model will learn the complex relationships and lead-lag dynamics between the constituent stocks and the overall market index.

To accomplish this forecasting task, we employ the Informer architecture; a Transformer variant tailored for long sequence time series forecasting (LSTF) [21]. Standard self-attention incurs $O(L^2)$ time and memory complexity and suffers from slow stepwise decoding [17]. Informer addresses these limitations via:

ProbSparse Self-Attention. To address the quadratic complexity of the standard self-attention mechanism, ProbSparse attention reduces the time and memory complexity to $O(L \log L)$. Instead of calculating attention scores for all query-key pairs, it selectively computes only the most significant ones. The importance of each query is determined by its query sparsity measurement, $M(q_i, K)$, which evaluates the similarity between the query's attention probability distribution and a uniform distribution. This "likeness" is measured using Kullback-Leibler divergence. Dropping a constant term, the final query sparsity measurement is defined as:

$$M(q_i, K) = \ln \sum_{j=1}^{L_K} \exp\left(\frac{q_i k_j^\top}{\sqrt{d}}\right) - \frac{1}{L_K} \sum_{j=1}^{L_K} \frac{q_i k_j^\top}{\sqrt{d}} \tag{4}$$

where q_i is the i-th query and K is the set of all keys. A larger $M(q_i, K)$ indicates a more diverse attention distribution, signifying an important query. Based on this measurement, only the top-u most dominant queries (where $u = c \cdot \ln L_Q$ for a sampling factor c) are kept. This results in a sparse query matrix $\bar{Q}$, and the final attention is calculated as:

$$\mathcal{A}(Q, K, V) = \text{Softmax}\left(\frac{\bar{Q}K^\top}{\sqrt{d}}\right) V \tag{5}$$

This allows each key to attend to only a few dominant queries, achieving significant computational savings while preserving performance.

Self-Attention Distillation. To handle extremely long input sequences without running into memory bottlenecks from stacking multiple encoder layers, Informer employs a self-attention distillation technique. After each ProbSparse self-attention layer in the encoder, this operation "distills" the most essential features and halves the sequence length. The procedure for moving from layer j to layer $j+1$ is defined as:

$$OFI_{j+1}^t = \text{MaxPool}\left(\text{ELU}\left(\text{Conv1d}\left(\left[OFI_j^t\right]_{AB}\right)\right)\right) \tag{6}$$

Here, $[\cdot]_{AB}$ refers to the output of the attention block, which is passed through a 1D convolutional filter, an ELU activation function, and finally a max-pooling layer with a stride of 2. This cascading reduction in input length drastically cuts down the total memory usage of the encoder to $O\big((2-\epsilon)L \log L\big)$, enabling the model to process longer inputs. To enhance robustness, the encoder also builds replicas on halved input sequences, creating a pyramid-like structure whose outputs are concatenated.

Generative Decoder. The traditional Transformer decoder predicts outputs in a step by step, auto-regressive manner, which is slow for long sequences. Informer's generative decoder predicts the entire output sequence in a single forward pass, significantly improving inference speed. This is achieved by feeding the decoder a specific input vector constructed as: $OFI_{de}^t = \text{Concat}\left(OFI_{\text{token}}^t, OFI_0^t\right)$. In this structure, OFI_{token}^t is a start token, which is a known segment of the sequence preceding the target prediction window. OFI_0^t is a placeholder for the target sequence, typically filled with zeros. By applying masked multi-head attention to this input, the decoder generates the entire output sequence in one forward step. This approach prevents each position from attending to subsequent positions and avoids the cumulative error propagation inherent in step by step decoding methods.

3.4 Stage 2: Spatio-Temporal Relationship Learning and Price Movement Classification

The second stage of our model is designed to interpret the predicted future market dynamics and classify the future direction of the market index. This stage takes as input the tensor of predicted future OFI sequences, $OFI_{\text{pred}} \in \mathbb{R}^{B \times N \times P \times C}$, which was generated by the Informer in the previous stage. Here, B is the batch size, N is the number of selected assets, P is the prediction sequence length (horizon), and C is the number of OFI features (e.g., 5 levels).

The goal of this stage is to learn the complex spatio-temporal interactions within this tensor. To this end, we design a Spatio-Temporal Classification Network that employs a hierarchical attention mechanism across the asset and temporal dimensions.

The core of our network is an attention module composed of L stacked *Stock-Time Attention* blocks. Each block is designed to explicitly model spatio-temporal dependencies through two sequential sublayers: a *Stock-wise Attention* and a *Time-wise Attention.* An initial linear projection maps the input feature dimension C to the model's hidden dimension D_{model}.

Stock-wise Attention. For each time step in the sequence, this sublayer captures the cross-sectional relationships among all N assets. Given the input features $OFI^l \in \mathbb{R}^{B \times N \times P \times D_{\text{model}}}$ from the l-th block, we first apply multi-head self-attention across the asset dimension (N). This is followed by a position-wise feed-forward network (FFN). The entire process incorporates residual connections and layer normalization. The computation is as follows:

$$A^l = \text{LayerNorm}\left(OFI^l + \text{MultiHeadStock}(OFI^l)\right) \tag{7}$$

$$OFI'^{l} = \text{LayerNorm}\left(A^{l} + \text{FFN}_{\text{stock}}(A^{l})\right) \tag{8}$$

Here, MultiHeadStock$(\cdot)$ performs multi-head attention on the dimension of stocks.

Time-wise Attention. Subsequently, this sublayer models the longitudinal, temporal patterns across the prediction horizon P for each individual asset. The intermediate output OFI'^{l} from the Stock-wise sublayer is passed through a Time-wise multi-head self-attention mechanism, followed by another FFN. This yields the final output OFI^{l+1} for the l-th block:

$$B^{l} = \text{LayerNorm}\left(OFI'^{l} + \text{MultiHeadTime}(OFI'^{l})\right) \tag{9}$$

$$OFI^{l+1} = \text{LayerNorm}\left(B^{l} + \text{FFN}_{\text{time}}(B^{l})\right) \tag{10}$$

where MultiHeadTime$(\cdot)$ applies multi-head attention along the time dimension (P). The output OFI^{l+1} is then fed into the next Stock-Time Attention block. Through these sequential operations, each block effectively integrates spatial (cross-asset) and temporal information, enriching the feature representation.

Prediction Head. The final feature map from the last block, $OFI^{L} \in \mathbb{R}^{B\times N\times P\times D_{\text{model}}}$, encapsulates the learned spatio-temporal dynamics and is fed into a prediction head. This head first flattens the asset (N) and time (P) dimensions of OFI^{L} to create a single sequence for each sample in the batch. An LSTM network then processes this flattened sequence to summarize its dynamics into a final hidden state, $h_{\text{final}} = \text{LSTM}(\text{Flatten}(OFI^{L}))$, which is a representation in $\mathbb{R}^{B\times D_{\text{model}}}$. Subsequently, this summarized state is passed to a linear classifier with dropout to generate the final predictions of the index's movement:

$$\text{Logits} = \text{Linear}(\text{Dropout}(h_{\text{final}})) \tag{11}$$

Training Objective. We train the entire model end-to-end using a multi-task learning approach. The total loss, $\mathcal{L}_{\text{total}}$, is a weighted sum of a forecasting loss and a classification loss, defined as:

$$\mathcal{L}_{\text{total}} = \lambda \cdot \mathcal{L}_{\text{forecast}} + (1-\lambda)\cdot \mathcal{L}_{\text{class}} \tag{12}$$

Here, $\mathcal{L}_{\text{forecast}}$ is the Mean Squared Error (MSE) between the predicted time series $\hat{Y}$ and the ground truth Y, while $\mathcal{L}_{\text{class}}$ is the Cross-Entropy loss between the predicted logits and the true class labels L. The hyperparameter $\lambda \in [0,1]$ balances the contribution of each task.

4 Experiment

4.1 Data

The dataset for this study consists of high-frequency LOB data from the Korea Exchange (KRX), spanning 30 trading days in 2022. The data comprises two main components: the KOSPI 200 index futures, which serves as our prediction target, and a set of input features derived from its constituents. From the 200 stocks in the KOSPI 200 index, we select 137 for which futures contracts exist. For each of these 137 stocks, data from both the individual stock futures and its corresponding spot asset are used as a paired input feature. This pool of 137 futures-spot pairs, along with the KOSPI 200 index futures data, forms the complete set of available inputs. To systematically evaluate the contribution of different market data, we constructed various dataset combinations for our experiments, as detailed in Table 2.

Prior to model training, the raw LOB data was preprocessed to ensure quality. We filtered the data to include only the official trading session (09:00-15:30 KST) and removed any records with invalid order book states. The 30 trading days of the resulting dataset were chronologically split into 20 days for training, 5 days for validation, and 5 days for testing.

A common characteristic of such financial data, when labeled for price direction prediction (i.e., upward, downward, stationary) as formally defined in 4.2, is a significant class imbalance. To address this issue during model training, we apply class weights to the loss function. A naive inverse frequency weighting can assign excessively large weights to minority classes, leading to training instability. To mitigate this effect and promote stable learning, we compute the weight w_c for each class c using a normalized inverse square root scheme, formulated as $w_c = (C/\sqrt{N_c})/\left(\sum_{i=1}^{C} 1/\sqrt{N_i}\right)$.

Here, N_c is the number of samples in class c, and C is the total number of classes. This formulation applies an inverse square root to the class frequencies, which dampens the weights assigned to rare classes, thus preventing instability. The weights are then normalized such that their sum equals the number of classes, maintaining a consistent scale for the overall loss.

For a rigorous and fair comparison, this class weighting scheme was consistently applied to all baseline models used in our experiments, including DeepLOB [20], BincTabel [15], MLPLOB [3], and TLOB [3]. This controlled experimental setup ensures that observed performance differences are attributable to the intrinsic merits of the model architectures rather than variations in the imbalance handling techniques.

4.2 Definition of Price Movement

The primary objective of this study is to predict short-term market direction. Accordingly, we define the target variable (label) as a categorical variable with three classes: 'Upward', 'Downward', and 'Stationary'. The label for each timestamp is determined based on future price movements derived from high-frequency LOB data. The generation process is as follows.

First, the raw tick data are preprocessed. To establish a robust price reference, we define the *mid-price* (P_{mid}) as the average of the best bid (P_{bid}) and best ask (P_{ask}) prices. This formulation serves to mitigate microstructure noise arising from the bid-ask bounce. The mid-price is calculated as:

$$P_{\text{mid}} = \frac{P_{\text{ask}} + P_{\text{bid}}}{2} \tag{13}$$

Since raw tick data arrives at irregular intervals, it is resampled into a fixed frequency time series. To this end, we construct a one-second time series $P_{\text{mid},n}$ by averaging the mid-prices of all ticks occurring within each interval $[t-h, t]$, where $h = 1s$. Let $N(t)$ be the cumulative number of ticks up to time t. The averaged mid-price at second t, denoted as $\bar{P}_t$, is formally computed as:

$$\bar{P}_t = \frac{1}{N(t) - N(t-h)} \sum_{n=N(t-h)+1}^{N(t)} P_{\text{mid},n} \tag{14}$$

Table 1: Model performance comparison at different prediction horizons. All metrics are reported as mean values.

Model	$k=5$			$k=10$			$k=30$			$k=60$		
	F1(macro)	Precision	Recall	F1(macro)	Precision	Recall	F1(macro)	Precision	Recall	F1(macro)	Precision	Recall
DeepLOB	0.3270	0.3134	0.3333	0.3171	0.3023	0.3333	0.2832	0.2461	0.3333	0.2195	0.1633	0.3334
BiNCTABL	0.3652	0.3647	0.4383	0.3783	0.3817	0.3858	0.3787	0.3806	0.3846	0.3737	0.3747	0.3741
MLPLOB	0.3766	**0.4158**	0.3709	0.3741	0.3817	0.3861	0.3733	0.3862	0.4002	0.3672	0.3700	0.3693
TLOB	0.3627	0.3634	**0.4561**	0.3776	0.3803	**0.4460**	0.3873	0.3912	0.4074	0.3802	0.3834	0.3878
Ours	**0.3871**	0.3971	0.3926	**0.4065**	**0.4124**	0.4073	**0.4238**	**0.4204**	**0.4299**	**0.3926**	**0.3929**	**0.3937**

For seconds with no trading activity, the last observed price is carried forward to create a continuous price series.

The label at any given time t is determined by the price change over a future horizon k. We consider multiple horizons, setting $k = 5, 10, 30$, and 60 seconds, to capture various short and mid-term market dynamics. The return R_t over horizon k is formulated as:

$$R_t = \frac{\bar{P}_{t+k} - \bar{P}_t}{\bar{P}_t} \tag{15}$$

Finally, the calculated return R_t is classified into a discrete label L_t based on a predefined threshold δ. The label is assigned as 'Upward' (2) for returns greater than or equal to δ, 'Downward' (0) for returns less than or equal to $-\delta$, and 'Stationary' (1) otherwise. For this study, we set a sensitive threshold of $\delta = 0.0002$ (2 basis points).

This choice is deliberately aligned with our data preprocessing methodology. By averaging tick data into one-second intervals, we apply a pre-smoothing filter that mitigates a significant portion of the high-frequency microstructure noise, such as bid-ask bounce. On this smoothed price series, even a minor deviation is more likely to represent a genuine, incipient trend rather than a random fluctuation. A sensitive threshold, therefore, enables our model to detect the very early stages of a directional price move.

We acknowledge that this threshold is below the typical breakeven point required to overcome transaction costs (e.g., commissions and slippage) in a live trading environment. However, the primary objective of this study is not to develop a directly profitable trading strategy, but rather to investigate the limits of statistical predictability in financial time series. By setting a sensitive threshold, we aim to capture subtle market inefficiencies and provide a more rigorous test of our model's ability to identify predictive patterns, even in small price movements.

4.3 Evaluation Metric

Our objective is to predict short-term market direction, a task for which the target labels are inherently imbalanced. The 'Stationary' class significantly outnumbers the 'Upward' and 'Downward' classes. In such a setting, accuracy is an inadequate metric, as a naive model that consistently predicts the majority class would achieve a high score without capturing the directional movements of interest.

To address this limitation, we adopt the F1-score (macro) as our primary evaluation metric. The F1-score (macro), defined as the harmonic mean of precision and recall, provides a more balanced assessment. It penalizes models that fail to perform well on the minority classes, making it a suitable measure of the model's ability to learn from imbalanced data in a comprehensive manner.

Table 2: Performance comparison with different input datasets (F1-score macro).

Dataset	$k=5$	$k=10$	$k=30$	$k=60$
KF	0.3431	0.3759	0.3957	0.3699
KF+CS	0.3704	0.3819	0.3966	0.3663
KF+CF	**0.3871**	**0.4065**	**0.4238**	**0.3926**

Abbreviations: KF = KOSPI 200 Futures only; CS = Constituent Spots; CF = Constituent Futures.

5 Results and Analysis

This section provides a multi-faceted evaluation of our proposed model. We first establish its superior predictive performance against established baselines across various prediction horizons ($k = 5, 10, 30$, and 60 seconds) using F1-score, as summarized in Table 1. We then examine this performance advantage through two targeted analyses: first, by investigating the predictive value of cross-asset order flow (Table 2), and second, by statistically validating our model's ability to identify primary market drivers through its attention mechanism (Table 3).

5.1 Predictive Value of Cross-Asset Order Flow Imbalance

Our approach diverges from previous work in two key aspects. First, we replace raw LOB snapshots with the OFI, a feature designed to capture immediate supply-demand pressures more effectively. Second, we aggregate OFI information from both the KOSPI 200 index futures and its constituent stock futures. This cross-asset perspective yields a consistent performance gain and empirically validates our central hypothesis: **that micro-level order flow from individual components carries material predictive power for the index as a whole**.

The performance advantage widens as the prediction horizon lengthens. At $k = 5$, the improvement is modest, suggesting that the index's own microstructure remains the dominant signal for ultra short-term forecasts. However, the performance gap grows substantially from $k = 10$ onward, with the model's F1-score peaking at $k = 30$ (0.4238). This result suggests that the model has identified a characteristic 30-second lag between predictive patterns emerging in constituent stocks and their subsequent impact on the aggregate index. Furthermore, the model retains a distinct edge at $k = 60$,

Table 3: Statistical comparison of DTW distances (Price Patterns) between attention ranked groups.

Date	Top-10	Bottom-10	Effect Size (Cohen's d)	Mann–Whitney U p-value
20221121	5.38 (±3.20)	8.36 (±3.78)	0.81	0.08
20221122	5.20 (±0.90)	7.85 (±1.94)	1.57	0.09
20221123	6.76 (±2.10)	9.93 (±2.46)	1.31	0.03
20221124	5.22 (±1.48)	7.44 (±3.14)	0.84	0.32
20221125	5.52 (±1.83)	7.53 (±2.25)	0.93	0.05

highlighting its ability to capture more persistent market dynamics beyond transient LOB noise.

These results strongly support a bottom-up methodology for short-term index forecasting. While single-asset models may suffice for instantaneous moves, incorporating disaggregated order flow markedly improves the robustness and accuracy of forecasts beyond a few seconds. Modeling the collective behavior of market components, therefore, provides a more holistic view of intraday market dynamics, offering a deeper understanding essential for sophisticated trading and risk management.

5.2 Analysis of Cross-Asset Information

To validate our central hypothesis that cross-asset relationships are critical for index prediction, we conducted an ablation study on the input datasets. Table 2 presents a performance comparison where we incrementally add information from constituent stocks to the baseline model, which uses only KOSPI 200 index futures (KF).

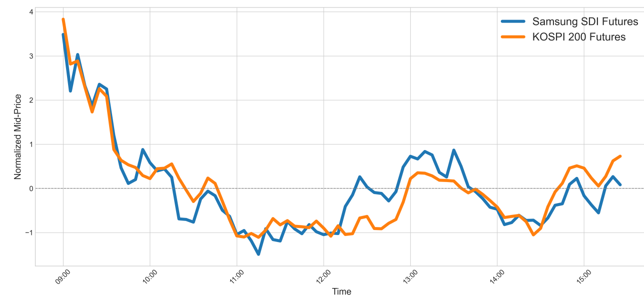
The results clearly demonstrate the value of incorporating information from constituent assets. Both models that include constituent data (KF+CS and KF+CF) consistently outperform the baseline KF model across nearly all prediction horizons. This performance validates our approach of modeling the market as an interconnected system and highlights the efficacy of the *Stock-wise Attention* mechanism in capturing valuable cross-sectional signals.

A more granular analysis reveals that the choice of constituent data, spot versus futures, is highly significant. The model using constituent futures (KF+CF) achieves the best performance across all horizons, and by a considerable margin at $k = 30$ and $k = 60$. This finding suggests that the KOSPI 200 index future exhibits a stronger and more immediate coupling with its constituent futures than with their underlying spot assets. This is an intuitive result from a financial perspective, as derivatives markets are often influenced by a similar set of participants and arbitrage-driven flows, leading to faster information transmission between them.

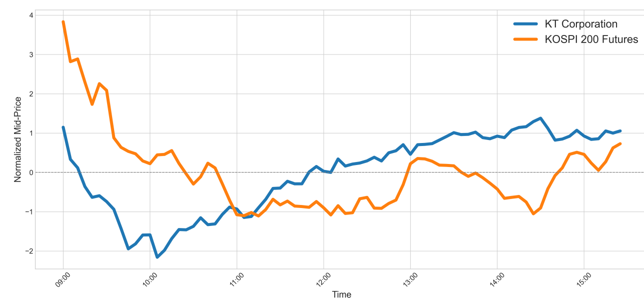
Interestingly, the predictive power of our best performing model (KF+CF) peaks at the $k = 30$ horizon. Performance remains strong but slightly decreases for the longer $k = 60$ horizon. This suggests a characteristic lead-lag time of approximately 30 seconds, where the predictive order flow patterns emerging in the constituent futures take time to be fully reflected in the index future's price movement. This insight underscores our model's ability not just to predict, but also to uncover subtle temporal dynamics within the market microstructure.

5.3 Validation of Primary Drivers via Price Pattern Similarity

The robust predictive performance established in the previous subsections lends credibility to our framework's interpretability. It suggests that the model's attention mechanism is not merely an



(a) Highest attention score (Samsung SDI Futures)



(b) Lowest attention score (KT Corporation Futures)

Figure 2: Price pattern comparison for the highest-ranked (a) and lowest-ranked (b) constituent futures against the KOSPI 200 index futures on a sample day (November 21, 2022).

arbitrary internal metric but reflects financially meaningful relationships. Building on the key finding that constituent futures (KF+CF) provide significantly more predictive power than constituent spots (KF+CS) (Table 2), we now seek to validate whether our model can identify the 'primary drivers' within this more informative dataset.

To achieve this, our analysis has a twofold objective: first, to identify a cohort of 'primary drivers' among the constituent futures based on the model's learned attention rankings; and second, to statistically verify whether the price movements of these identified drivers are indeed more similar to the index's price movements compared to less influential futures. We ranked all constituent stocks by their average received attention score and designated the top 10 as the 'Top-10 Group' and the bottom 10 as the 'Bottom-10 Group.' We then employed Dynamic Time Warping (DTW) [2] to measure the similarity between the mid-price time series of each stock and the index futures. DTW is a robust algorithm for comparing time series that may be out of phase, where a lower distance signifies higher pattern similarity. To rigorously compare the two groups, we used two statistical metrics. First, we calculate Cohen's d [4]

to measure the effect size, which quantifies the magnitude of the difference between the groups. By convention, a Cohen's d value of 0.2 is considered a 'small' effect, 0.5 a 'medium' effect, and 0.8 a 'large' effect. This is crucial for evaluating practical significance, especially in low power settings with small sample sizes. Second, we applied the Mann-Whitney U test [12], a non-parametric test ideal for small samples as it does not assume a normal distribution, to assess statistical significance (p-value).

The results, summarized in Table 3, provide strong support for our hypothesis. Across all five test days, the Top-10 Group, as identified by our model's attention, exhibited a consistently and substantially lower mean DTW distance to the index than the Bottom-10 Group. The practical magnitude of this difference, measured by Cohen's d, was confirmed to be large to very large across all five days (d values ranging from 0.81 to 1.57). The Mann-Whitney U test confirmed that this difference was statistically significant ($p < 0.05$) on one day ($p = 0.03$) and reached borderline significance on another ($p = 0.05$). The lack of consistent statistical significance on the remaining days is likely attributable to the low statistical power inherent in the small sample size (N=10 per group).

To visually supplement this statistical analysis, Figure 2 provides a qualitative example from a sample day (November 21, 2022). It compares the normalized mid-price movements of the constituent future that received the highest attention score (Samsung SDI Futures) and the one that received the lowest (KT Corporation Futures) against the KOSPI 200 index futures. As is evident, the price pattern of the Samsung SDI Futures shows a strong visual resemblance to the index's movement. In contrast, the KT Corporation Futures exhibits considerable divergence, often moving in a dissimilar direction. This visual evidence corroborates our quantitative findings, demonstrating that the model's learned attention effectively identifies primary drivers whose price dynamics are synchronized with the aggregate index.

In conclusion, this analysis provides strong validation for the interpretability of our framework. The constituent futures identified as primary drivers via high attention are empirically shown to have price dynamics more similar to the aggregate index, as measured by their consistently lower DTW distances. Although statistical significance at the conventional level $p < 0.05$ was not achieved on every test day due to the small sample size, the practical magnitude of the difference was consistently large or very large, as confirmed by the Cohen's d effect sizes. This robust separation between the top and bottom attention groups suggests that the model's learned attention is not an arbitrary internal metric but a financially relevant mechanism. Our framework, therefore, demonstrates a valuable ability to autonomously identify key market drivers and quantitatively confirm their relationship with the aggregate market.

6 Conclusion

In this study, we proposed a novel, interpretable deep learning framework that models the market as an interconnected system to predict index movements, overcoming the limitations of single-asset, black box approaches. Experiments on KOSPI 200 data show our model significantly outperforms existing baselines. We empirically demonstrate that cross-asset order flow from constituent futures is a more potent predictor than from spot data, and identify a characteristic 30-second lead-lag dynamic where the model's performance peaks.

Crucially, we validate the model's interpretability. The attention mechanism effectively identifies primary market drivers, as confirmed by the quantitative similarity of their price patterns to the aggregate index. Ultimately, our work presents not just an accurate predictor, but a transparent analytical tool providing a holistic understanding of market microstructure, offering valuable insights for both academic research and sophisticated financial applications.

This study has room for improvements. First, high quality, high-frequency LOB data essential for model training is difficult to access and costly, restricting this study's scope to a specific market and time period. Verifying the model's generalizability across diverse market conditions remains for future research. Second, the threshold (δ) used for labeling price movements does not account for real-world transaction costs. A threshold sufficient to offset such costs would create a severe class imbalance problem: making the 'Up' and 'Down' classes too infrequent, thereby hindering the model's ability to train effectively. Consequently, this study aims primarily at detecting statistically significant micro-price movements rather than achieving direct trading profitability.

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